

Binomial expansion - Set 2

TMUA practice questions by topic. Specification MM2.4. 4 questions.

Each question has one correct answer. No calculator. Answers and worked solutions are in the separate solutions PDF.

- 1 The expansion of $\left(2x - \frac{1}{x^2}\right)^9$ contains a term in x^k for each of which of the following values of k ?
- I** $k = 0$
 - II** $k = 3$
 - III** $k = -6$
- A** none of them
- B** I only
- C** II only
- D** III only
- E** I and II only
- F** I and III only
- G** II and III only
- H** I, II and III
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- 2 The coefficient of x^2 in $(1+x)^n + (1+x)^{n+1}$ is 36. Find n .
- A** 4
- B** 8
- C** 6
- D** 5
- E** 9
- F** 7

3 n is a positive integer. Which of the following statements is/are true?

I The coefficient of x^n in $(1+x)^{2n}$ is $\binom{2n}{n}$.

II The sum of the coefficients in the expansion of $(1-x)^n$ is 0.

III $(1+x)^n \geq 1+nx$ for all $x \geq 0$.

A none of them

B I only

C II only

D III only

E I and II only

F I and III only

G II and III only

H I, II and III

4 How many terms in the expansion of $(x+y+z)^4$ (after collecting like terms) are there?

A 16

B 15

C 10

D 5

E 12

F 81